



**Politécnico
Castelo Branco**

Escola Superior
de Tecnologia

School of Technology
Bachelor's Degree in Computer Engineering

Cloud Computing

Mini Project Assignment

Project objectives

Create a small **original** web application based on the Microsoft Azure cloud computing platform, with the following mandatory requirements:

- Must use GitHub as a code repository, integrated with Azure App Service.
- The project's cloud infrastructure must be able to be created automatically, using a script or template.
- The application must use the NoSQL CosmosDB database.
- The application must use at least one Docker Container.
- The application must use at least one Azure Function (serverless computing).
- The application must use at least one cloud storage service, for example BLOB storage (or other).

The use of other additional services will be valued.

AI tools may be used, but the report must describe which tools were used and for what purpose.

There will be three submissions throughout the semester, which are described on the following page.

Group composition

The mini-project must be carried out by groups of two or three students, whose members are chosen by each group.

Assessment #1 – due 27 February (2 points)

Materials to be submitted: first version of the report in PDF format, with a description of the idea, the objectives of that idea, the Azure services that will be used, and how each of these services will be used. The report must also contain a Gantt chart with the work plan and the division of tasks among the group members.

Assessment: the assessment will be based on the originality and feasibility of the idea, how the various services will be used, and the report.

Assessment #2 – due 17 April (4 points)

Materials to be submitted: second version of the report in PDF format, now also containing a description of the status of the mini-project implementation. At this stage, some of the application's features are expected to be up and running.

Assessment: the assessment will be based on the work done up to this point, the report, and the students' performance in presenting and defending their work.

Assessment #3 – due on 22 May (14 points)

Materials to be submitted: final version of the report in PDF format, describing the complete implementation of the mini-project. All project files must also be submitted, including the script or template that allows the creation of the infrastructure to be automated.

At this stage, all application features are expected to be working, using all mandatory cloud services. The report should contain a detailed description of the mini-project implementation and also:

- A forecast of the monthly cost of the solution.
- A theoretical study on how this solution could be implemented using European cloud companies (which ones should be indicated) instead of Azure.

Assessment: assessment will be based on the work carried out, the report and the students' performance in the presentation and defence of the work.

Report

The report must use IEEE-style citations/references. AI tools may be used in the report, provided they are cited as the source of the information.